NPTEL ONLINE CERTIFICATION COURSES



Course Name: Software Project Management

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TOPIC: Project Evaluation and Programme Management



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CONCEPTS COVERED

- Business case for a project
- Project portfolio management
- Project evaluation



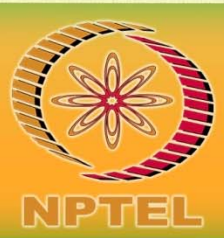
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The business case

- Feasibility study is also called as “business case”
- Main focus is to determine whether it would be financially & technically feasible to develop the S/W.
- Provides a justification for starting the project
- Should show that the benefits of the project will exceed the costs of development, implementation and operation (or production)
- Needs to take account of business risks





Types of feasibility study

- Technical Feasibility
- Financial Feasibility
- Operational Feasibility





Contents of a business case / feasibility study

1. Introduction/ background
2. The proposed project
3. The market
4. Organizational and operational infrastructure
5. The benefits
6. Outline implementation plan
7. Costs
8. The financial case
9. Risks
10. Management plan

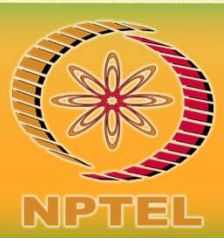




Contents of a business case cont…

- **Introduction/background:** describes a problem to be solved or an opportunity to be exploited
- **The proposed project:** a brief outline of the project scope
- **The market:** the project could be to develop a new product (e.g. a new computer game). The likely demand for the product would need to be assessed.





Contents of a business case cont…

- **Organizational and operational infrastructure:** How the organization would need to change. This would be important where a new information system application was being introduced.
- **Benefits** Normally, these should be expressed in financial terms. In the end, it is up to the client to assess these – as they are going to pay for the project.





Contents of a business case cont…

- **Outline implementation plan:** How the project is going to be implemented. This should consider the disruption to an organization that a project might cause.
- **Costs:** The implementation plan will supply information to establish these.
- **Financial analysis:** Combines costs and benefit data to establish value of the project

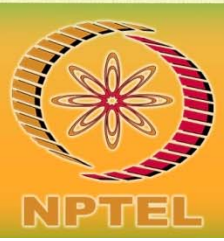




Contents of a business case cont…

- **Risks:** Two types
- ✓ **Project Risk:** related to threats to successful project execution.
- ✓ **Business Risk:** related to factors threatening the benefits of the delivered project.
- In business case, main focus is on Business Risk.
- **Management Plan:** Detailed plan for smooth management of the organization.





Project Portfolio Management (PPM)

- Provides an overview of all the projects that an organization is undertaking.
- Prioritizes the allocation of resources to projects.
- Decides which new projects should be accepted and which existing ones should be dropped.





Project portfolio management cont …

The concerns of project portfolio management include:

- Evaluating proposals for projects
- Assessing the risk involved with projects
- Deciding how to share resources between projects
- Taking account of dependencies between projects
- Removing duplication between projects
- Ensuring that necessary developments have not been inadvertently missed.





Project portfolio management cont…

There are three elements to PPM:

1. **Project portfolio definition**
   - Create a central record of all projects within an organization
   - Must decide whether to have ALL projects in the repository or, say, only ICT projects
   - Note difference between new product development (NPD) projects and renewal projects e.g. for process improvement





Project portfolio management cont…

2. **Project portfolio management**

   Actual costing and performance of projects can be recorded and assessed

3. **Project portfolio optimization**

   Information gathered above can be used to achieve better balance of projects e.g. some that are risky but potentially very valuable balanced by less risky but less valuable projects





Pros and cons of Project portfolio management

Pros:

- Allows some small ad-hoc tasks to be done outside the portfolio e.g. quick fixes to systems.

Cons:

- While the full-time staff are allocated to a project, they may effectively be part-time because, they still have routine work to do.
- The official project portfolio may not accurately reflect organizational activity, if some projects are excluded.





Evaluation of individual projects

Technical assessment:

- Consists of evaluating whether the required functionality can be achieved with current affordable technologies.
- Cost of the technology adopted must be taken into account in the financial assessment (cost-benefit analysis).

Evaluation of individual projects cont...

Financial assessment (Cost benefit analysis):

This relates to an individual project. You need to:

- Identify all the costs which could be:
  - Development costs (e.g. Salary to staffs)
  - Set-up cost (e.g. Hardware cost, recruitment and staff training cost)
  - Operational cost (e.g. Cost for operating the system after installation)
- Identify the value of benefits
- Check that benefits are greater than costs





Summary

- Discussed the concept of business case and its contents.
- Presented the concerns of project portfolio management (PPM) and the elements to PPM.
- In order to evaluate individual projects, one has to carry out
  - ✓ Technical Assessment
  - ✓Financial Assessment (C-B Analysis)





References :

1. B. Hughes, M. Cotterell, R. Mall, *Software Project Management*, Sixth Edition, McGraw Hill Education (India) Pvt. Ltd., 2018.
2. E. M. Awad, *Systems Analysis and Design*, Second Edition. Galgotia Publications Pvt. Ltd., 2009.
3. R. Mall, *Fundamentals of Software Engineering*, Fifth Edition, PHI Learning Pvt. Ltd., 2018.






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